

■ Design Google Drive / Dropbox

System Design Interview | Media Streaming / File Storage

algomaster.io | Deep Dive Notes

■ Key Requirements

Functional	Upload/download files; Sync across devices; Share files/folders; Version history; Offline access; Conflict resolution
Non-Functional	1B users; Files from 1KB to 50GB; Fast sync (<5s); Strong consistency for same user; Deduplication to save storage
Scale	Google Drive: 15M TB stored; Dropbox: 500M users; Block-level sync for efficiency

■ High-Level Architecture

- Client splits files into fixed-size blocks (4MB); upload only changed blocks (delta sync)
- Block Storage Service: each block stored in S3 addressed by content hash (SHA-256); automatic deduplication
- Metadata Service: tracks files, folder structure, block list per version; PostgreSQL with file_id + version
- Sync Service: long-poll or WebSocket; notifies clients of changes; sends diff (changed block list)

■ Block-Level Sync Deep Dive

- **Chunking:** File split into 4MB blocks; each block identified by SHA-256 hash; store blocks in content-addressable storage
- **Delta Sync:** On file change: compare block list before/after; upload only changed/new blocks; reduces bandwidth by 90%+
- **Deduplication:** Same block hash = same content; stored once, referenced by multiple files/users; saves ~50% storage in practice
- **Upload Flow:** Client → pre-check which blocks server already has → upload missing blocks to S3 → update metadata DB with new block list + version
- **Download/Sync:** Server notifies client of changed file → send new block list → client downloads only missing blocks → reassemble file

■ Data Model

- files: file_id, owner_id, name, parent_folder_id, is_deleted, current_version_id — PostgreSQL
- file_versions: version_id, file_id, size, created_at, block_ids[] — stores ordered list of block hashes
- blocks: block_hash(PK), s3_path, size, ref_count — content-addressable; ref_count for garbage collection
- shares: file_id, shared_with_user_id, permission(view/edit), expires_at — access control

■ Conflict Resolution

- Same-user, different device: last-write-wins based on timestamp; show both versions if meaningful conflict
- Dropbox approach: create 'conflicted copy' with username suffix; user resolves manually

- OT (Operational Transformation): for Google Docs-style real-time collaboration; transform concurrent ops
- CRDT approach: mathematically conflict-free but complex; better for structured data (rich text)

■ Sync Notification

Long Polling	WebSocket
✓ Simpler implementation	✓ True real-time push
✓ Works through all firewalls	✓ Lower latency
✓ Client re-polls after response	✓ Persistent connection overhead
✓ Dropbox uses this	✓ Google Drive uses this

■ Security & Permissions

- Encryption at rest: AES-256 per file; encryption at transit: TLS 1.3
- Pre-signed URLs: client gets time-limited S3 URL for direct upload/download; reduces proxy bandwidth
- Access control: owner can share with view/edit/comment permissions; folder permissions cascade to children
- Audit log: every file access/modification logged for compliance (GDPR, HIPAA); immutable append-only log